FINITENESS, BIRATIONALITY AND THE REDUCED CASE FOR THE NORMALIZATION OF A LOCALLY ALGEBRAIC SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Parts (2) and (3) are discharged by the author with the single sentences 'Part (2) is Morphisms, Lemma ...' and 'Part (3) is Morphisms, Lemma ...'; these sentences are reproduced verbatim, and they are navigation pointers rather than arguments, exactly as the author wrote them.

Editorial note. The counted spans carry 9 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-normalization-locally-algebraic is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Locally finite type schemes over a field are locally Noetherian and universally Nagata. Morphisms, Lemmas morphisms-lemma-finite-type-noetherian and morphisms-lemma-ubiquity-nagata, invoked at proof lines 4921-4923 and 4931-4934. Supplies the applicability of the normalization definition and the finiteness of the normalization morphism.
- Normalization is a disjoint union of normal integral schemes with dominant morphism. Morphisms, Lemma morphisms-lemma-normalization-normal, invoked at proof lines 4928-4930. Supplies the structural part of (1).
- Normalization of a reduced scheme, and birationality for reduced algebraic schemes. Morphisms, Lemmas morphisms-lemma-normalization-reduced and morphisms-lemma-normalization-birational, invoked at proof lines 4939 and 4942. Supplies parts (2) and (3) as direct references.
- A quasi-compact open of a locally Noetherian scheme has finitely many irreducible components. Properties, Lemma properties-lemma-Noetherian-irreducible-components, invoked at proof lines 4924-4926. Supplies that the normalization definition applies.

Source reference. Original definition 035N: sources/stacks-chow/tags.txt, source lines 4134–4134. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035S: sources/stacks-chow/tags.txt, source lines 4139–4139. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BXC: sources/stacks-chow/tags.txt, source lines 14975–14975. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035Q: sources/stacks-chow/tags.txt, source lines 4137–4137. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035O: sources/stacks-chow/tags.txt, source lines 4135–4135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035B: sources/stacks-chow/tags.txt, source lines 4122–4122. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BA8: sources/stacks-chow/tags.txt, source lines 14201–14201. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic scheme over k. Let $\nu : X^\nu \to X$ be the normalization morphism, see Morphisms, Definition 035N. Then*

1. *ν is finite, dominant, and X^ν is a disjoint union of normal irreducible locally algebraic schemes over k,*
2. *ν factors as $X^\nu \to X_{red} \to X$ and the first morphism is the normalization morphism of X_{red},*
3. *if X is a reduced algebraic scheme, then ν is birational,*
4. *if X is a variety, then X^ν is a variety and ν is a finite birational morphism of varieties.*

Proof. Since X is locally of finite type over a field, we see that X is locally Noetherian (Morphisms, Lemma 01T6) hence every quasi-compact open has finitely many irreducible components (Properties, Lemma 0BA8). Thus Morphisms, Definition 035N applies. The normalization X^ν is always a disjoint union of normal integral schemes and the normalization morphism ν is always dominant, see Morphisms, Lemma 035Q. Since X is universally Nagata (Morphisms, Lemma 035B) we see that ν is finite (Morphisms, Lemma 035S). Hence X^ν is locally algebraic too. At this point we have proved (1).

Part (2) is Morphisms, Lemma 035O.

Part (3) is Morphisms, Lemma 0BXC.

Part (4) follows from (1), (2), (3), and the fact that X^ν is separated as a scheme finite over a separated scheme. □